

EXPT NO:2	LOAD AND VISUALIZE A DATASET USING TABLEAU and PowerBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS) 1.

1. Why is Tableau a preferred tool for data visualization?

Tableau is a powerful data visualization tool with an easy drag-and-drop interface. It allows users to create interactive charts and dashboards quickly. It supports multiple data sources and handles large datasets efficiently. Users can analyze data without extensive programming knowledge. It is widely used for business intelligence and decision-making.

2. What is the primary use of a line chart in data analysis?

A line chart is mainly used to show trends over time. It helps visualize increases or decreases in data values. It is useful for identifying seasonal patterns and growth. Multiple data series can be compared in one chart. It is commonly used for sales, stock, and performance analysis.

3. When would you use a tree map instead of a bar chart?

A tree map is used to display hierarchical data with many categories. It shows both category size and proportion in a compact layout. It is useful when screen space is limited. Tree maps help compare multiple sub-categories at once. They provide a quick overview of data distribution.

4. What is the significance of filters in Tableau visualizations?

Filters allow users to view only the required portion of the data. They make dashboards more interactive and user-friendly. Filters help compare different categories, regions, or time periods. They reduce unnecessary information from the visualization. This improves analysis and decision-making.

5. How does Tableau handle geographic data for visualizations?

Tableau automatically recognizes geographic fields like Country, State, and City. It assigns geographic coordinates to these fields. Users can create maps without manually entering locations. Geographic data can be colored and sized using measures like sales. This helps analyze location-based business performance.

IN-LAB EXERCISE:

OBJECTIVE:

To load a dataset into Tableau, create a variety of visualizations, and derive insights from the data using different chart types available in Tableau.

PROCEDURE:

Dataset Selection:

Use the Global Superstore Sales Dataset (available on Kaggle or Tableau's sample datasets).

Ensure the dataset includes fields like Order Date, Category, Region, Sales, Profit, and Customer Segment.

Data Loading:

Open Tableau and connect to the dataset.

Review the dataset structure, field data types, and missing values.

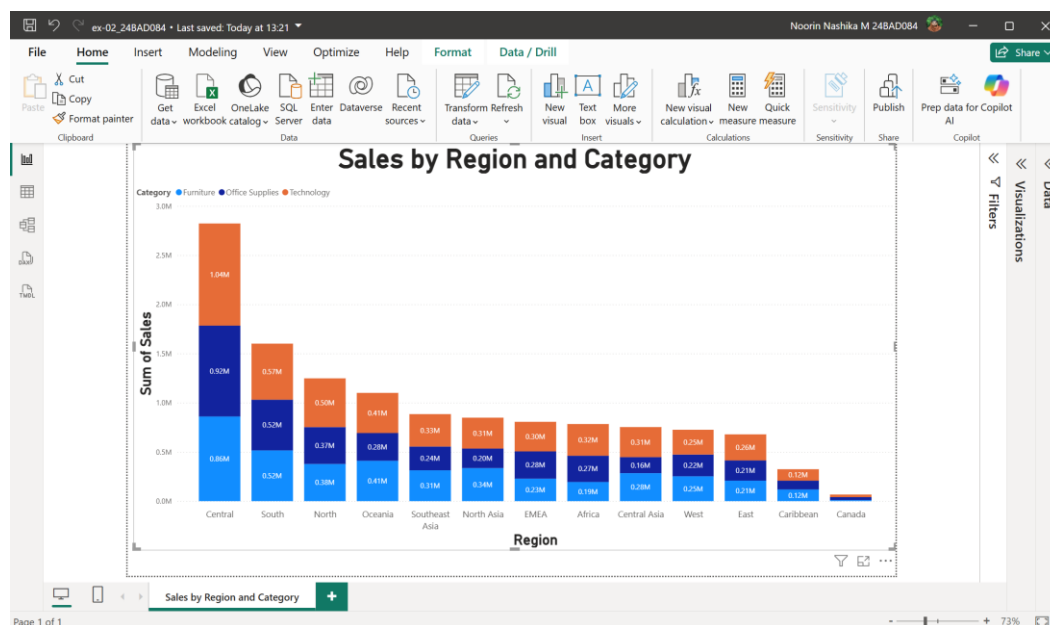
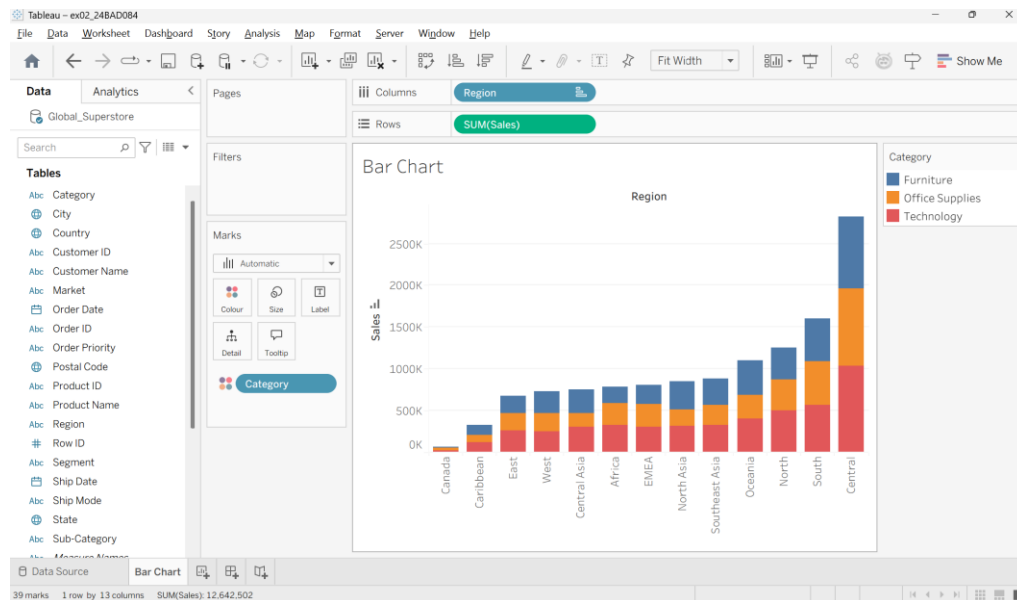
Chart Creations (Step-by-Step):

a) Bar Chart:

Objective: Compare total sales by region and category.

Plot Region on the x-axis and Sales on the y-axis.

Add Category as a color dimension.

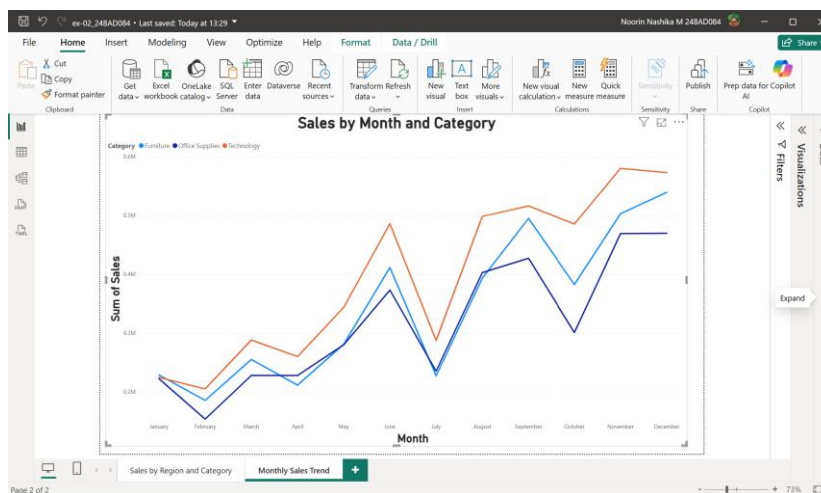
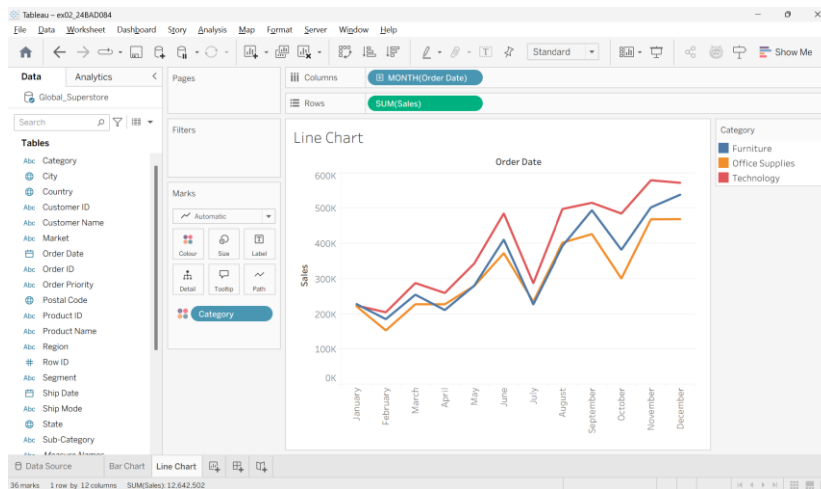


b) Line Chart:

Objective: Visualize sales trends over time.

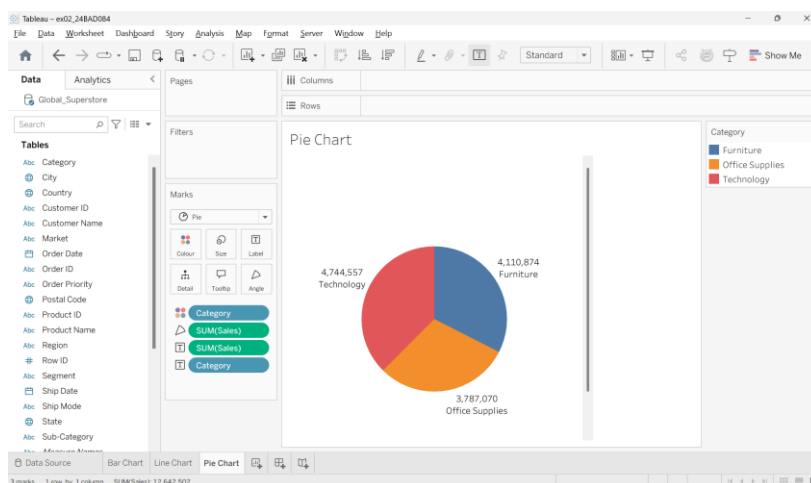
Place Order Date on the x-axis and Sales on the y-axis.

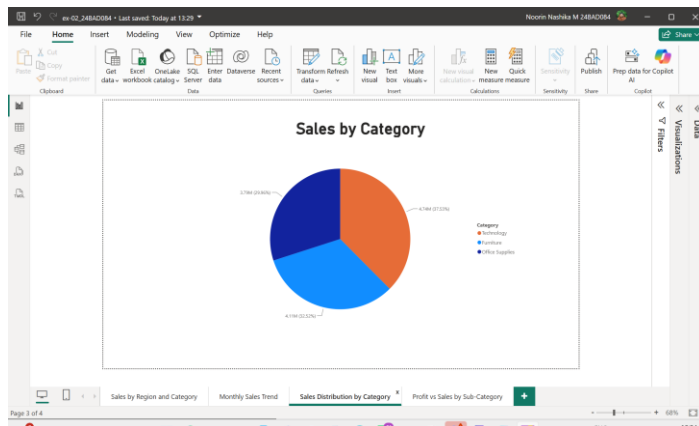
Set granularity to Month and color by Category.



c) Pie Chart:

Objective: Show sales distribution by product category. Assign Category as the dimension and Sales as the measure.



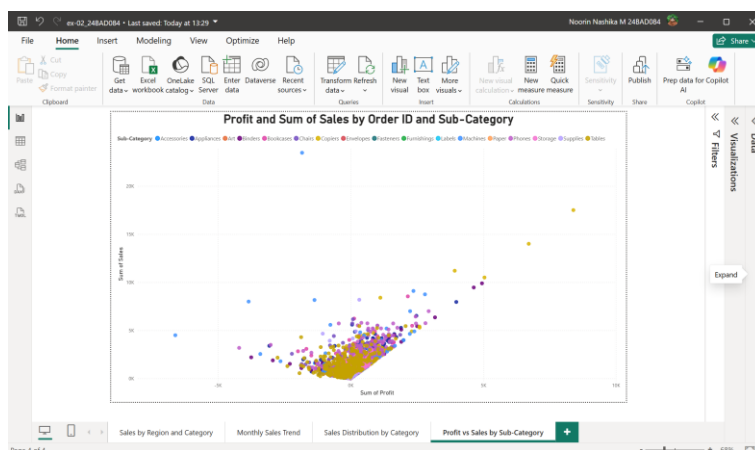
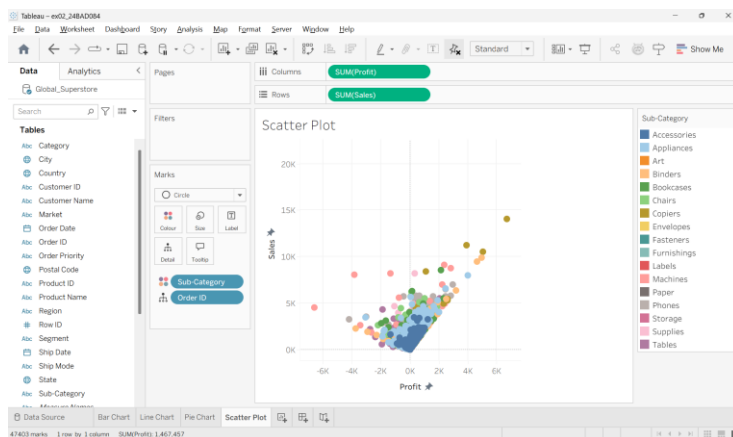


d) Scatter Plot:

Objective: Examine the relationship between profit and sales.

Plot Profit on the x-axis and Sales on the y-axis.

Use Sub-Category as the color dimension.



e) Heatmap:

Objective: Compare sales across categories and regions.

Use Region and Category as dimensions. Use

Sales as the measure and apply color coding.

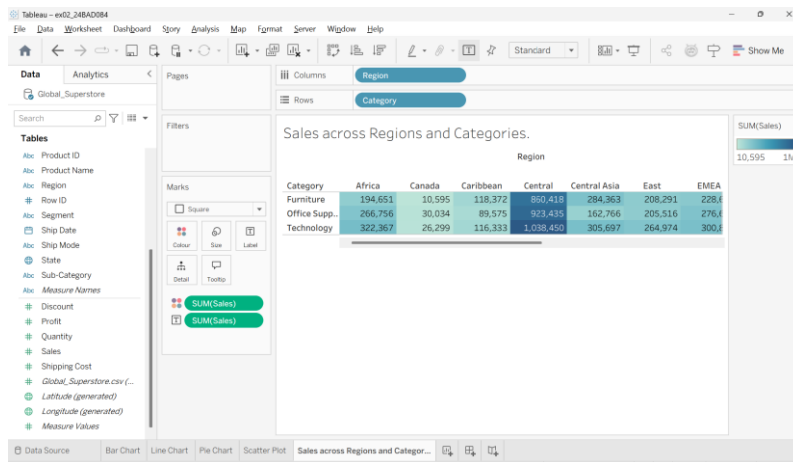


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Visualizations: Sales Heatmap by Region and Category

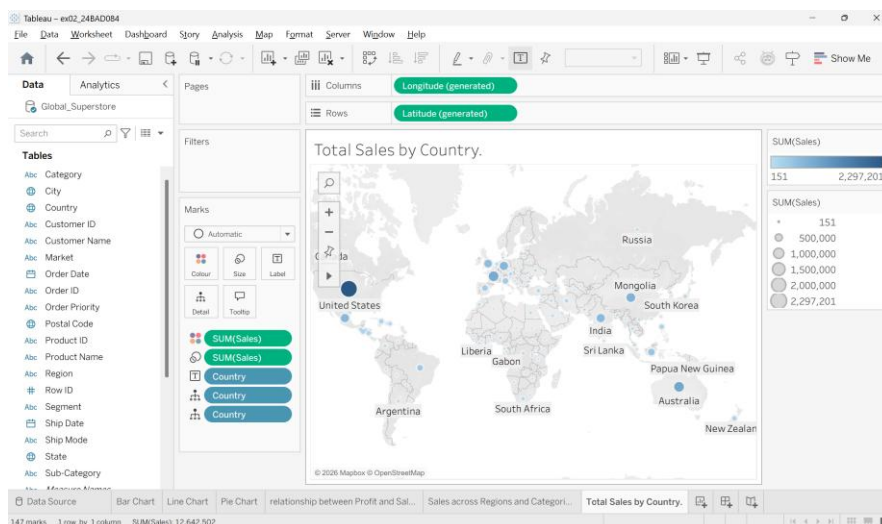
Region	Furniture	Office Supplies	Technology	Total
Africa	194,650.64	266,755.53	322,367.04	783,773.21
Canada	10,595.28	30,034.08	26,298.81	66,928.17
Caribbean	118,372.40	89,575.42	116,333.05	324,280.86
Central	860,417.58	923,435.28	1,038,449.66	2,822,302.52
Central Asia	284,362.90	162,766.35	305,697.32	752,826.57
East	208,291.20	205,516.05	264,973.98	678,781.24
EMEA	228,620.74	276,685.99	300,854.58	806,161.31
North	377,629.92	374,733.45	495,802.23	1,248,165.60
North Asia	335,716.46	198,554.77	314,038.55	848,309.78
Oceania	410,468.00	281,713.63	408,002.98	1,100,184.61
South	515,749.62	515,161.34	569,996.09	1,600,907.04
Southeast Asia	313,386.70	241,285.08	329,751.38	884,423.17
West	252,612.74	220,853.25	251,991.83	725,457.82
Total	4,110,874.19	3,787,070.23	4,744,557.50	12,642,501.91

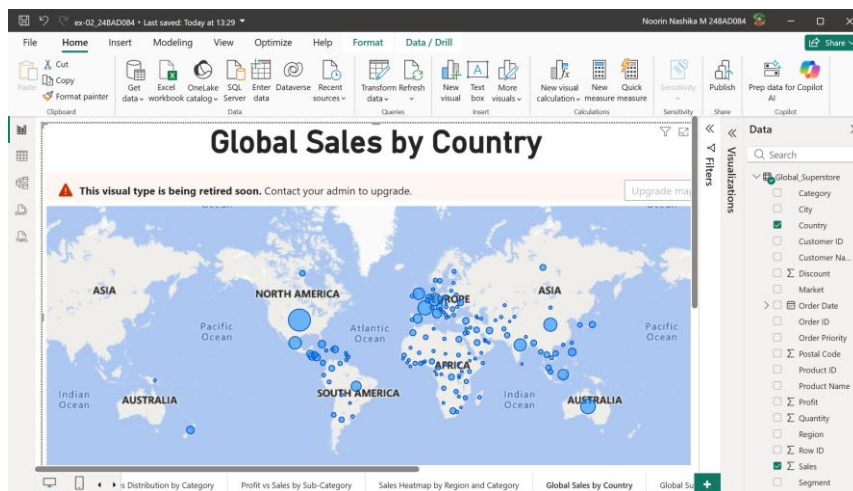
f) Geographical Map:

Objective: Visualize total sales by country.

Set Country as the geographical field. Use

Sales to define size and color gradients.

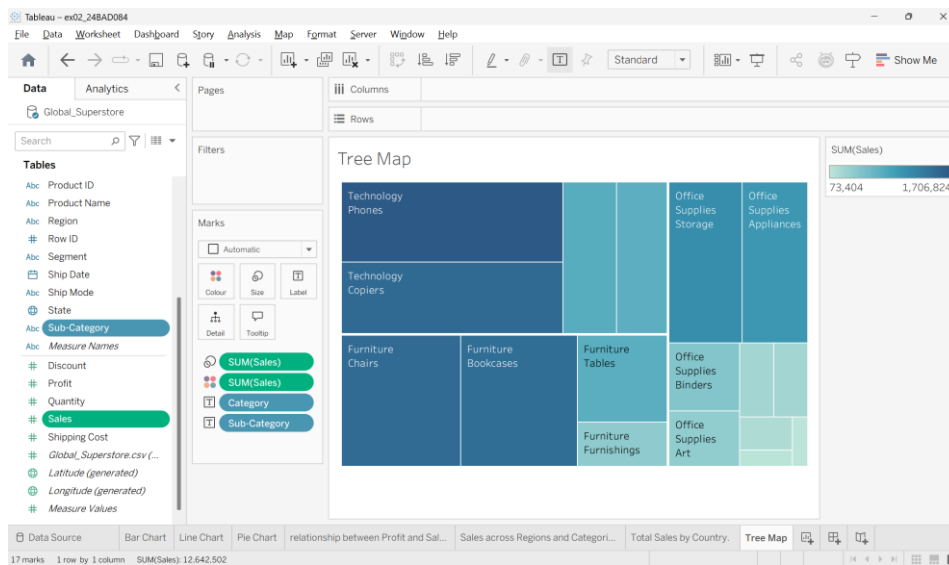




g) Tree Map:

Objective: Display sales distribution by category and sub-category.

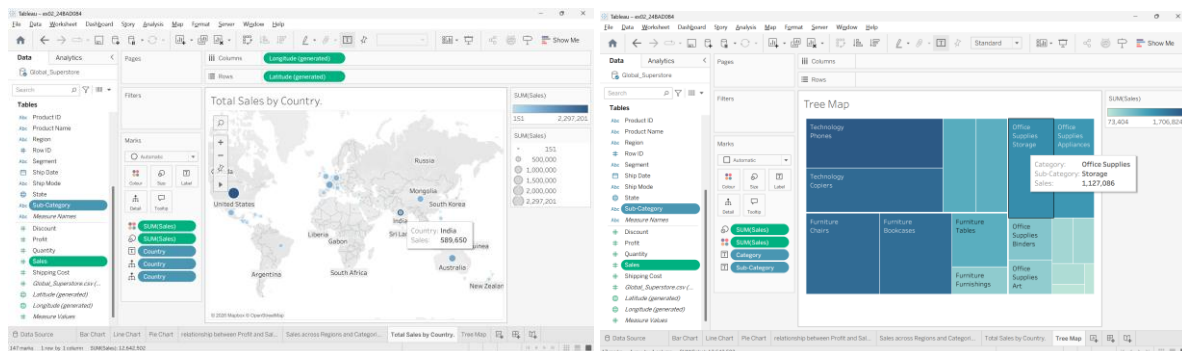
Use Category and Sub-Category as dimensions, and Sales as the measure.



Customization and Interactivity:

Add filters to explore data subsets interactively (e.g., by year, region, or category).

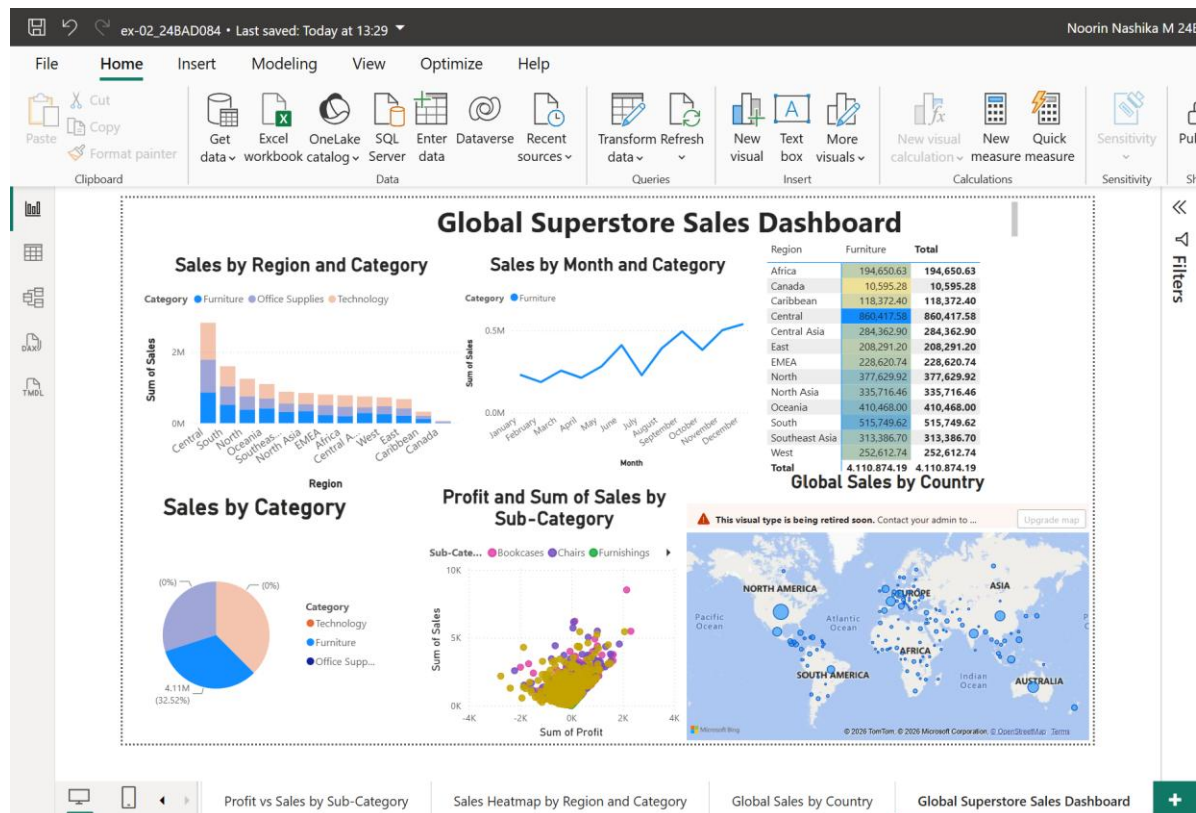
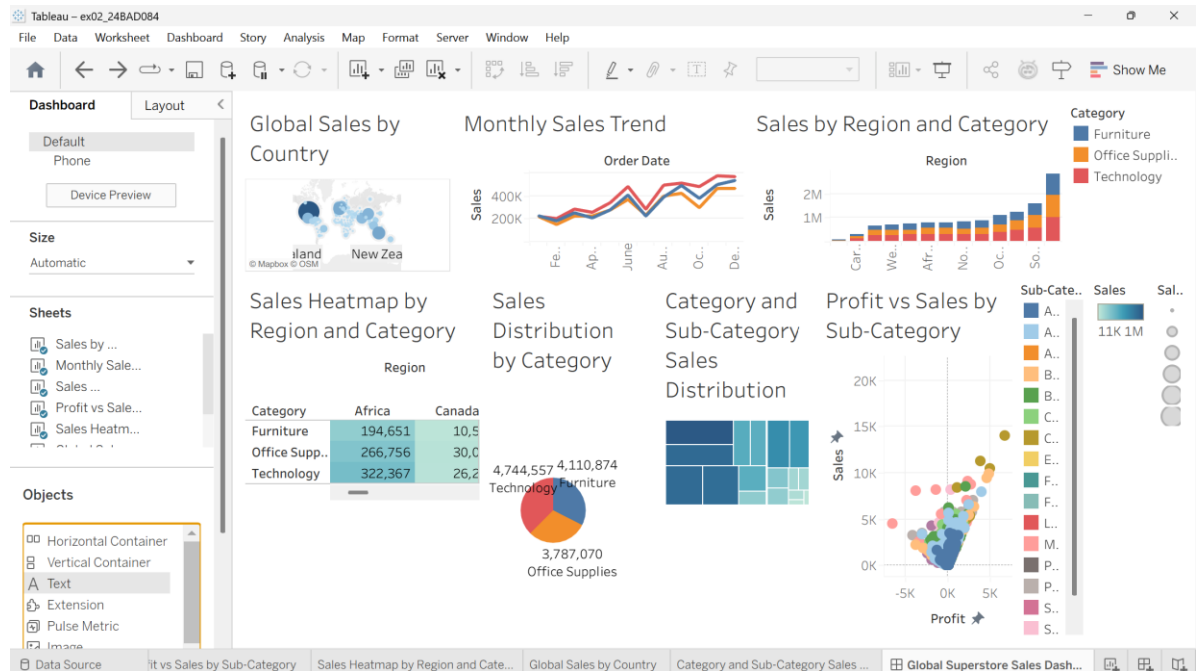
Include annotations, labels, and formatting for clarity.



Dashboard Creation:

Combine all charts into a single dashboard to present a cohesive story.

Add interactive filters and tooltips for dynamic exploration.



Experiment Dataset:

Global Superstore Sales Dataset

Source: <https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Fields Used:

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Order Date

Region

Category

Sub-Category

Sales

Profit

Country

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

Bar Chart

1. What regions had the highest and lowest total sales?

The West region recorded the highest total sales. The Central region had comparatively lower sales. Sales varied across different regions based on customer demand. Regional comparison helps identify high-performing markets. This information supports better business planning.

2. How does the performance of categories vary across regions?

Technology performed well in most regions. Furniture showed moderate sales performance. Office Supplies had consistent but comparatively lower sales. Category performance differed depending on the region. Such analysis helps identify regional preferences.

Line Chart

3. Identify the month with peak sales.

November recorded the highest sales among all months. Sales increased significantly during the holiday season. Seasonal demand contributed to the growth. Other months showed moderate performance. The trend indicates yearly sales fluctuations.

4. Which category shows consistent growth in sales over time?

Technology showed the most consistent growth in sales. Customer demand increased steadily over time. Furniture and Office Supplies had moderate fluctuations. Technology maintained strong performance throughout the period. This indicates higher market demand.

Pie Chart

5. Which category contributes the most to overall sales?

Technology contributed the highest share of total sales. It occupied the largest portion of the pie chart. Furniture and Office Supplies contributed smaller shares. The chart clearly shows category-wise distribution. Technology remained the leading category.

6. Are there any categories with negligible sales contributions?

No category had negligible sales. All categories contributed to the overall revenue. However, Furniture had a relatively smaller share. Office Supplies showed moderate contribution. Each category played an important role in total sales.

Scatter Plot

7. Is there a noticeable correlation between profit and sales?

A positive correlation exists between profit and sales. Higher sales generally resulted in higher profits. Some products showed high sales with low profit. A few points represented losses despite good sales. This helps identify less profitable products.

8. Which sub-category shows the highest sales but low profit?

Tables showed high sales with relatively low profit. This indicates lower profit margins. High discounts may have reduced profitability. Other sub-categories performed better financially. Such analysis supports pricing decisions.

Heatmap

9. Which region-category combination has the highest sales?

The West region with Technology category showed the highest sales. This combination stood out clearly in the heatmap. Other regions had moderate performance. Color intensity helped identify top-performing combinations. It provides an easy visual comparison.

10. Identify any regions with poor sales performance across all categories.

The Central region showed comparatively lower sales. Most categories performed below the leading regions. Demand was relatively weaker in this area. The heatmap highlighted these differences. Such insights help improve regional strategies.

Geographical Map

11. Which country has the highest total sales?

The United States recorded the highest total sales. It appeared with the largest marker on the map. Other countries had comparatively lower sales. Geographic visualization makes regional comparison easier. It helps identify important markets.

12. Are there regions with minimal or no sales?

Yes, a few regions showed very low sales. Some locations had limited customer activity. These areas appeared with smaller markers. Such regions may require marketing improvements. Geographic maps highlight these differences effectively.

Tree Map

13. Which sub-category dominates the sales in each category?

Phones dominated the Technology category. Chairs had strong sales in Furniture. Storage performed well in Office Supplies. The tree map displayed the size of each sub-category. Larger blocks represented higher sales.

14. Are there any sub-categories with unexpectedly low sales?

Fasteners showed relatively low sales. Labels also had a smaller contribution. These sub-categories occupied less space in the tree map. Lower sales indicate limited customer demand. This helps identify weaker product segments.

Dashboard

15. What insights can you derive when combining all visualizations?

The dashboard provides a complete view of sales performance. It combines trends, regional analysis, category comparison, and profitability. Users can identify patterns quickly. Interactive charts improve understanding of the data. It supports better business decisions.

16. How do filters enhance the utility of the dashboard?

Filters allow users to analyze specific regions, categories, or time periods. They make the dashboard interactive and flexible. Users can focus on relevant information. This improves the speed of analysis. Filters make dashboards more useful for decision-making.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		